Confidence-Aware, Explainable Multi-Species Bird Presence Detection System Using Bioacoustic Signals

Technical Report

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Date: February 2026

Abstract

This project presents a deep learning-based system for automatic bird species detection from audio recordings. The system addresses the challenge of manually analyzing long environmental recordings by providing automated species identification with confidence scores and visual explanations.

The system uses mel-spectrogram representation of audio signals and employs an EfficientNet-B0 convolutional neural network for classification. The model was trained on 1,926 recordings from the Xeno-Canto database, covering 54 North American bird species.

Key Results: • Top-1 Accuracy: 72.26% • Top-5 Accuracy: 85.55% • F1 Score (Weighted): 72.26% • Real-world validation: Correctly identified unseen recordings

The system features Gradient-weighted Class Activation Mapping (Grad-CAM) for explainability, showing which parts of the spectrogram influenced the model's decision. A web-based interface built with Streamlit allows researchers to upload recordings, search for target species, and generate PDF reports.

Keywords: Bioacoustics, Bird Detection, Deep Learning, EfficientNet, Grad-CAM, Explainable AI, Convolutional Neural Networks, Audio Classification

1. Introduction

1.1 Background

Bird species monitoring is crucial for ecological research, conservation efforts, and biodiversity assessment. Traditional methods rely on expert ornithologists manually listening to recordings, which presents several challenges:

- Time-consuming: Hours of recordings must be analyzed for minutes of actual bird calls
- Expensive: Requires trained experts with specialized knowledge
- Subjective: Different experts may disagree on species identification
- Limited scalability: Cannot process large datasets efficiently

Automated acoustic monitoring using machine learning offers a solution by processing recordings quickly and consistently. Recent advances in deep learning, particularly convolutional neural networks (CNNs), have shown remarkable success in audio classification tasks when combined with spectrogram representations.

1.2 Problem Statement

A field researcher collects audio recordings in a forest environment. Each recording typically contains: • Background noise (wind, insects, rain, human activity) • Potentially multiple bird species calling simultaneously • Variable recording quality depending on equipment and conditions

The researcher needs to answer several key questions: 1. Is my target species present in this recording? (Binary: Yes/No) 2. How confident is the detection? (Probability score) 3. Where in the recording does the call occur? (Temporal localization) 4. Why does the system think this? (Explainable evidence)

Current commercial solutions like BirdNET and Merlin provide species identification but lack explainability features that allow researchers to verify and understand the basis for predictions.

1.3 Objectives

This project aims to build a comprehensive system that:

1. Automatically detects bird species from audio recordings with high accuracy 2. Provides calibrated confidence scores for each detection 3. Localizes detections temporally within the recording 4. Explains predictions through visual heatmaps (Grad-CAM) 5. Offers an intuitive web interface accessible to non-technical researchers 6. Generates professional PDF reports for documentation 7. Provides uncertainty warnings when predictions may be unreliable

1.4 Scope

The scope of this project includes: • Dataset: Xeno-Canto recordings (Creative Commons licensed) • Species coverage: 54 North American bird species • Input formats: WAV, MP3, OGG, FLAC audio files • Output: Species identification with confidence scores, temporal localization, and visual explanations • Deployment: Web-based interface using Streamlit framework

2. Methodology

2.1 System Architecture

The complete system pipeline consists of the following stages:

1. Audio Input: Users upload recordings in various formats (WAV, MP3, OGG, FLAC) 2. Preprocessing: Audio is resampled to 22,050 Hz, converted to mono, and split into 5-second chunks with 50% overlap 3. Feature Extraction: Each chunk is converted to a mel-spectrogram representation 4. Classification: EfficientNet-B0 neural network processes spectrograms and outputs class probabilities 5. Explainability: Grad-CAM generates attention heatmaps showing important regions 6. Aggregation: Chunk-level predictions are aggregated to recording-level results 7. Output: Species predictions with confidence scores, timelines, and visual explanations

FIGURE 1

System Architecture

[Pipeline diagram showing: Audio Input → Mel-Spectrogram → EfficientNet → Predictions + Grad-CAM]

Figure 1: System Architecture

2.2 Dataset

The dataset was sourced from Xeno-Canto (xeno-canto.org), a citizen science project hosting bird sound recordings from around the world under Creative Commons licenses.

Dataset Statistics: • Total recordings downloaded: 4,521 • Usable recordings (with complete metadata): 1,926 • Species covered: 54 North American bird species • Total audio duration: Approximately 40 hours • Geographic focus: North America

Data was split as follows: • Training set: 70% (1,338 recordings, 23,839 chunks) • Validation set: 15% (287 recordings, 5,568 chunks) • Test set: 15% (287 recordings, 5,404 chunks)

The split was performed at the recording level (not chunk level) to prevent data leakage, with stratification by species to ensure balanced representation across splits.

Metric	Value
Total recordings downloaded	4,521
Usable recordings	1,926
Species covered	54
Total audio chunks	34,811
Training chunks	23,839 (70%)
Validation chunks	5,568 (15%)
Test chunks	5,404 (15%)

Table 1: Dataset Statistics

2.3 Audio Preprocessing

Audio preprocessing involved three main steps:

Step 1: Standardization • Sample rate: 22,050 Hz (sufficient to capture frequencies up to 11 kHz, covering the typical bird call range of 1-10 kHz) • Channels: Converted to mono • Bit depth: 16-bit PCM

Step 2: Chunking • Duration: 5 seconds per chunk (long enough to capture complete bird calls) • Overlap: 50% (2.5-second hop) to ensure calls at chunk boundaries are not missed • Result: 34,811 total chunks from 1,926 recordings

Step 3: Quality Filtering • Silent chunks removed (RMS amplitude < 0.001) • Corrupted files identified and excluded • File integrity verified

2.4 Feature Extraction

Audio waveforms were converted to mel-spectrograms using the Short-Time Fourier Transform (STFT) with mel-scale frequency mapping.

The mel scale is a perceptual scale that approximates human hearing, providing higher resolution at lower frequencies where bird calls contain important distinguishing features.

Mathematical Process: 1. STFT: Apply windowed Fourier transform to extract frequency content over time 2. Power Spectrum: Compute magnitude squared of STFT output 3. Mel Filterbank: Apply triangular mel-scale filters to compress frequency representation 4. Log Compression: Convert to decibel scale to normalize dynamic range 5. Normalization: Scale values to 0-1 range for neural network input

Spectrogram Parameters: • n_mels: 128 frequency bins • n_fft: 2048 samples (approximately 93ms window at 22,050 Hz) • hop_length: 512 samples (approximately 23ms between frames) • fmin: 150 Hz (below most bird calls) • fmax: 15,000 Hz (above most bird calls) • Output shape: (128, 216) per 5-second chunk

Parameter	Value	Purpose
n_mels	128	Frequency bins
n_fft	2048	FFT window (~93ms)
hop_length	512	Frame hop (~23ms)
fmin	150 Hz	Min frequency
fmax	15,000 Hz	Max frequency

Table 2: Spectrogram Parameters

FIGURE 2

Mel-Spectrogram Conversion

[Audio waveform (left) converted to mel-spectrogram (right)]

Figure 2: Mel-Spectrogram Conversion

2.5 Model Architecture

The classification model is based on EfficientNet-B0, a state-of-the-art convolutional neural network architecture developed by Google Research.

Rationale for EfficientNet: • Achieves excellent accuracy with fewer parameters than alternatives • Pretrained on ImageNet enables transfer learning • Compound scaling balances network depth, width, and resolution • Mobile Inverted Bottleneck Convolution (MBConv) blocks provide efficiency • Squeeze-and-Excitation (SE) attention mechanism highlights important features

Architecture Details: • Input: (3, 128, 216) - 3-channel spectrogram (replicated for RGB compatibility) • Backbone: EfficientNet-B0 pretrained on ImageNet • Feature dimension: 1,280 after global average pooling • Classifier: Dropout (0.3) → Dense (54 classes) → Softmax • Total parameters: Approximately 4 million • All parameters trainable (full fine-tuning)

2.6 Training Configuration

The model was trained using the following configuration:

Optimizer: AdamW • Learning rate: 0.001 with cosine annealing warm restarts • Weight decay: 0.0001 (L2 regularization) • Beta parameters: β■ = 0.9, β■ = 0.999

Loss Function: Cross-Entropy with Label Smoothing • Label smoothing factor: 0.1 • Prevents overconfident predictions

Learning Rate Schedule: Cosine Annealing with Warm Restarts • T■ = 10 epochs • T_mult = 2 • Minimum learning rate: 1e-6

Regularization Techniques: • Dropout: 0.3 in classifier layer • Label smoothing: 0.1 • Weight decay: 0.0001 • Early stopping: Patience of 7 epochs

Data Augmentation (Applied Online During Training): • Time masking: Random 0-30 frames set to zero • Frequency masking: Random 0-20 mel bands set to zero • Gaussian noise: Added with σ = 0.01

Training Environment: • GPU: NVIDIA GeForce RTX 3050 (4GB VRAM) • Mixed precision: FP16 (Automatic Mixed Precision) • Batch size: 16 • Total training time: Approximately 2 hours • Epochs: 30 (early stopped at epoch 17) • Best model saved at epoch 10

Parameter	Value
Optimizer	AdamW
Learning Rate	0.001
Batch Size	16

Epochs	30 (stopped at 17)
Loss	CrossEntropy + Label Smoothing
GPU	RTX 3050 (4GB)

Table 3: Training Configuration

2.7 Explainability: Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) was implemented to provide visual explanations for model predictions.

Algorithm: 1. Forward pass: Compute feature maps from the last convolutional layer and obtain prediction 2. Backward pass: Compute gradients of the target class score with respect to feature maps 3. Weight calculation: Global average pool the gradients to obtain importance weights 4. Heatmap generation: Compute weighted combination of feature maps 5. ReLU activation: Keep only positive contributions 6. Upsampling: Resize heatmap to match input spectrogram dimensions

Mathematical Formulation: • Importance weights: $\alpha_k = (1/Z) \times \Sigma\blacksquare\ \Sigma\blacksquare\ (\partial y^c / \partial A^k_{ij})$ • Grad-CAM heatmap: $L^c = \mathrm{ReLU}(\Sigma_k \alpha_k \times A^k)$

Where: • y^c is the score for target class c • A^k is the activation map of channel k • Z is the number of pixels in the activation map

Interpretation: • Red/Yellow regions indicate high model attention (likely bird call) • Blue regions indicate low attention (background noise) • Researchers can verify if the model focuses on relevant acoustic features

3. Results

3.1 Training Performance

The model was trained for 17 epochs before early stopping was triggered. Training progress showed rapid initial learning followed by gradual improvement.

Training History Summary: • Epoch 1: Train Accuracy 45.23%, Validation Accuracy 38.52% • Epoch 5: Train Accuracy 96.86%, Validation Accuracy 64.42% • Epoch 10: Train Accuracy 99.83%, Validation Accuracy 68.79% (Best Model) • Epoch 17: Train Accuracy 98.25%, Validation Accuracy 62.79% (Early Stopping)

The gap between training accuracy (99.83%) and validation accuracy (68.79%) indicates some degree of overfitting, which is discussed in the limitations section.

Epoch	Train Acc	Val Acc	Notes
1	45.23%	38.52%	Initial
5	96.86%	64.42%	Rapid learning
10	99.83%	68.79%	Best Model
17	98.25%	62.79%	Early Stop

Table 4: Training History

FIGURE 3

Training Curves

[Train/Val accuracy and loss over epochs]

Figure 3: Training Curves

3.2 Test Set Evaluation

The final model was evaluated on the held-out test set of 5,404 chunks from 287 recordings.

Overall Performance Metrics: • Top-1 Accuracy: 72.26% • Top-3 Accuracy: 82.83% • Top-5 Accuracy: 85.55% • F1 Score (Macro): 63.21% • F1 Score (Weighted): 72.26%

The Top-5 accuracy of 85.55% indicates that the correct species is among the top 5 predictions in the vast majority of cases, which is practically useful for researchers who can quickly verify from a short list.

Metric	Value
Top-1 Accuracy	72.26%

Top-3 Accuracy	82.83%
Top-5 Accuracy	85.55%
F1 (Weighted)	72.26%

Table 5: Test Performance

FIGURE 4

Confusion Matrix

[54×54 confusion matrix heatmap]

Figure 4: Confusion Matrix

3.3 Per-Species Performance

Performance varied across species, with some achieving over 90% F1 score while others performed less well.

Top 5 Performing Species: 1. Selasphorus platycercus (Broad-tailed Hummingbird): F1 = 0.92 2. Botaurus lentiginosus (American Bittern): F1 = 0.91 3. Coccyzus erythropthalmus (Black-billed Cuckoo): F1 = 0.91 4. Cyanocitta cristata (Blue Jay): F1 = 0.91 5. Anthus rubescens (American Pipit): F1 = 0.88

Species with distinctive, consistent vocalizations tended to perform better, while species with variable calls or limited training samples showed lower performance.

3.4 Explainability Results

Grad-CAM visualizations demonstrate where the model focuses when making predictions, enabling researchers to verify that predictions are based on relevant acoustic features.

FIGURE 5

Grad-CAM: Correct Prediction

[Spectrogram + Heatmap + Overlay for correct classification]

Figure 5: Grad-CAM: Correct Prediction

FIGURE 6

Grad-CAM: Incorrect Prediction

[Spectrogram + Heatmap + Overlay for misclassification]

Figure 6: Grad-CAM: Incorrect Prediction

3.5 Real-World Validation

To validate real-world performance, the system was tested on recordings not present in the training data.

Real-World Test Results: • Recording XC475302: Actual = Blue-winged Warbler, Predicted = Blue-winged Warbler (97.0% confidence) ✓ • Recording XC416747: Actual = Blue-winged Warbler, Predicted = Blue-winged Warbler (96.6% confidence) ✓ • Recording XC481834: Actual = Blue-winged Warbler, Predicted = Bald Eagle (95.3% confidence) ✗

Real-world accuracy: 66.7% (2 out of 3 correct)

The misclassification of XC481834 with high confidence (95.3%) highlights an important limitation: the model can be "confidently wrong." This motivated the implementation of uncertainty warnings and the 80% confidence threshold for positive detection.

Recording	Actual	Predicted	Conf.	Result
XC475302	Blue-winged Warbler	Blue-winged Warbler	97%	✓
XC416747	Blue-winged Warbler	Blue-winged Warbler	96.6%	✓
XC481834	Blue-winged Warbler	Bald Eagle	95.3%	✗

Table 6: Real-World Tests

3.6 Deployed Application

A web interface was developed using Streamlit, providing audio upload, species search, visualization, and PDF report generation capabilities accessible to non-technical researchers.

FIGURE 7

Streamlit Application

[Screenshot of web interface]

Figure 7: Streamlit Application

FIGURE 8

PDF Report Output

[Generated PDF report example]

Figure 8: PDF Report Output

4. Discussion

4.1 Strengths

The developed system demonstrates several strengths:

1. Reasonable Accuracy: Achieving 72.26% top-1 and 85.55% top-5 accuracy on a 54-class problem with limited training data is competitive with similar systems in the literature.

2. Explainability: The Grad-CAM visualizations provide valuable insight into model decision-making, allowing researchers to verify that predictions are based on relevant acoustic features rather than artifacts or noise.

3. Confidence Awareness: The 80% threshold for positive detection and the reliability scoring system help users understand when predictions can be trusted.

4. Practical Interface: The Streamlit web application makes the system accessible to researchers without programming expertise.

5. Real-world Validation: Testing on completely unseen recordings demonstrated that the model can generalize beyond the training distribution, correctly identifying 2 out of 3 test cases.

6. Professional Output: The PDF report generation feature provides documentation suitable for research publications and field reports.

4.2 Limitations

Several limitations should be acknowledged:

1. Overfitting: The 31% gap between training accuracy (99.83%) and validation accuracy (68.79%) indicates the model has memorized some training-specific patterns. This likely results from: • Limited training data (only 1,926 recordings) • High model capacity (4 million parameters) • Insufficient data augmentation

2. Confidence Calibration: The model can predict with very high confidence (>95%) on incorrect species, as demonstrated by the XC481834 misclassification. This occurs because: • Softmax outputs tend to be overconfident • The model has not been exposed to sufficient diversity in recording conditions

3. Single-label Classification: The current system predicts only one species per 5-second chunk, but real forest recordings often contain multiple species calling simultaneously.

4. Incomplete Dataset: Only 46% of downloaded recordings (1,926 out of 4,521) were usable due to metadata scraping failures, potentially limiting the diversity of training examples.

5. Class Imbalance: Training data ranges from 98 to 1,253 samples per species (12.8x ratio), which may bias predictions toward more common species.

6. Recording Variability: The model may perform differently on recordings made with different equipment, at different distances, or in different environmental conditions than the training data.

5. Future Work

Several directions for future improvement have been identified:

Short-term Improvements: • Complete metadata scraping for remaining 2,205 recordings to expand training data • Implement stronger data augmentation (SpecAugment, Mixup, CutMix) • Apply confidence calibration using temperature scaling or Platt scaling • Add more uncertainty quantification metrics

Medium-term Improvements: • Convert to multi-label classification to detect multiple species per chunk • Implement transfer learning from BirdNET embeddings for improved features • Build ensemble models combining multiple architectures for robustness • Add species range validation using geographic metadata

Long-term Vision: • Develop real-time detection capability for streaming audio input • Create mobile applications for iOS and Android field deployment • Implement active learning with user feedback for continuous improvement • Integrate with camera systems for multi-modal bird identification

6. Conclusion

This project successfully developed a confidence-aware, explainable bird species detection system using bioacoustic signals and deep learning.

Key Contributions:

1. Working End-to-End System: A complete pipeline from audio input to species prediction with visual explanation, accessible through an intuitive web interface.

2. Explainability: Grad-CAM visualizations show which spectrogram regions influence predictions, enabling researchers to verify and understand model decisions.

3. Confidence Awareness: The 80% threshold for positive detection and reliability scoring system help users understand prediction trustworthiness, addressing the problem of overconfident incorrect predictions.

4. Practical Validation: Real-world testing on unseen recordings demonstrated both the capabilities and limitations of the system, providing honest assessment of performance.

Key Performance Metrics: • Top-1 Accuracy: 72.26% • Top-5 Accuracy: 85.55% • Species Covered: 54 • Real-world Tests: 2/3 Correct (66.7%)

The system serves as a valuable tool for researchers needing quick, explainable bird species identification from audio recordings. While acknowledging its limitations through uncertainty warnings, it provides a practical solution for bioacoustic monitoring that can be improved with additional data and continued development.

The honest acknowledgment of limitations, particularly the potential for high-confidence misclassifications, represents an important contribution toward responsible deployment of AI systems in scientific applications where trust and verification are essential.

Metric	Value
Top-1 Accuracy	72.26%
Top-5 Accuracy	85.55%
Species Covered	54
Real-world Tests	2/3 (66.7%)

Table 7: Summary Metrics

7. References

[1] Tan, M., & Le, Q. (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. International Conference on Machine Learning (ICML).

[2] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. IEEE International Conference on Computer Vision (ICCV).

[3] Kahl, S., Wood, C. M., Eibl, M., & Klinck, H. (2021). BirdNET: A deep learning solution for avian diversity monitoring. Ecological Informatics, 61, 101236.

[4] Kong, Q., Cao, Y., Iqbal, T., Wang, Y., Wang, W., & Plumbley, M. D. (2020). PANNs: Large-Scale Pretrained Audio Neural Networks for Audio Pattern Recognition. IEEE/ACM Transactions on Audio, Speech, and Language Processing, 28, 2880-2894.

[5] McFee, B., Raffel, C., Liang, D., Ellis, D. P., McVicar, M., Battenberg, E., & Nieto, O. (2015). librosa: Audio and Music Signal Analysis in Python. Proceedings of the 14th Python in Science Conference, 18-25.

[6] Xeno-Canto Foundation. (2024). Xeno-Canto: Sharing Bird Sounds from Around the World. https://xeno-canto.org

[7] Park, D. S., Chan, W., Zhang, Y., Chiu, C. C., Zoph, B., Cubuk, E. D., & Le, Q. V. (2019). SpecAugment: A Simple Data Augmentation Method for Automatic Speech Recognition. Interspeech 2019.

[8] Loshchilov, I., & Hutter, F. (2019). Decoupled Weight Decay Regularization. International Conference on Learning Representations (ICLR).

8. Appendix

A. Species List (Sample - First 20)

#	Scientific Name	English Name	Samples
1	Myiarchus cinerascens	Ash-throated Flycatc	1253
2	Spizella breweri	Brewer's Sparrow	1249
3	Spinus tristis	American Goldfinch	1062
4	Artemisiospiza belli	Bell's Sparrow	895
5	Polioptila caerulea	Blue-gray Gnatcatche	827
6	Setophaga striata	Blackpoll Warbler	778
7	Calypte anna	Anna's Hummingbird	720
8	Empidonax alnorum	Alder Flycatcher	711
9	Dolichonyx oryzivorus	Bobolink	706
10	Toxostoma rufum	Brown Thrasher	695
11	Spizelloides arborea	American Tree Sparro	585
12	Thryomanes bewickii	Bewick's Wren	574
13	Mniotilta varia	Black-and-white Warb	552
14	Corvus brachyrhynchos	American Crow	549
15	Setophaga virens	Black-throated Green	517
16	Scolopax minor	American Woodcock	513
17	Setophaga ruticilla	American Redstart	499
18	Riparia riparia	Bank Swallow	497
19	Turdus migratorius	American Robin	482
20	Selasphorus platycercus	Broad-tailed Humming	454

Table A1: Species List (Sample)

B. Software Stack

Component	Technology

Deep Learning	PyTorch 2.10, timm
Audio Processing	librosa 0.10
Web Interface	Streamlit
PDF Generation	ReportLab

Table B1: Software